From Resistance to Recovery: Breaking the Metabolic Lock

Coco · Integration Medicine III · A conventional central-ADH path vs B proposed renal / mineral path

A — The Conventional Trap: Treating the Symptom, Not the Cause

Severe polydipsia
PU / PD

Hyposthenuria
USG 1.004*

Old specialist label
PRIMARY CENTRAL DI (CDI)

? Metabolic data missed
Intact PTH 25.00 pmol/L (2.5× high)* · ionized Ca normal*
Lecture-note values. Not on the Antech / IDEXX pages in the student packet.

Vicious cycle after the label

Vetoryl (trilostane)
Stopped after anorexia and lethargy recorded → **Cessation of meds** → **Desmopressin cycle**
1 drop TID → 2 drops BID
Printed chart. Failure to concentrate.

Agitation and night waking
Owner sees “dementia,” not dilute urine

Vomiting / GI upset
Later bronchitis and pneumonia

What the printed chart actually supports

A CDI label was entered without an ACTH, LDDS, or water-deprivation page in the packet.
Three things moved in two weeks: trilostane stopped, a conflicting UTI treated,
and desmopressin started. A response is not proof of primary central ADH deficiency.
Creatinine stayed 0.3–0.8 mg/dL. Normal creatinine does not close polyuria.
USG 1.004* is a lecture-note figure. Printed in-house values start at 1.006 / 1.008.

B — The Metabolic Working Model: Unlocking V2 / AQP2

Working diagnosis (proposed)
Secondary nephrogenic DI · secondary hyperparathyroidism

Proposed lock · excess PTH* · inactive V2 / AQP2

DDAVP (key) — PTH* holds — V2 / AQP2 inactive

USG 1.004* · failure to concentrate
The key does not turn while the lock is held.

Hypothesis only: PTH 2.5× high* is proposed to resist the V2 / AQP2 cascade. No receptor assay is in the record.

Targeted therapeutic plan (as taught)

Dietary correction
Ca : P ratio · less bovine tissue

Calcitriol / vitamin D3
Confirm the number in Plumb

Goal: suppress PTH into the reference interval, then retest ionized Ca, PTH, phosphorus, and 25-OH D.

Proposed unlock · corrected minerals · active V2 / AQP2

DDAVP (key) → V2 / AQP2 active

USG 1.012–1.017* · sleep improved*
Corrected minerals are proposed to release the cascade.

How a doctor should read this panel

A rising USG after diet and vitamin D support is compatible with a metabolic contribution.
It does not by itself prove receptor-level resistance, and it does not exclude residual
CDI, hyperadrenocorticism, or infection. Sort every claim into four boxes:
observed finding · interpretation · causal hypothesis · diagnostic conclusion.
This figure is a teaching model. It is not a proof that Coco did not have CDI.

* Lecture-note values (PTH 25.00 pmol/L; USG 1.004–1.006 and 1.012–1.017). Confirm against the laboratory page before publication. Dr. Yujin Kim, DVM, PhD, FFCP · Lewyt College of Veterinary Medicine.